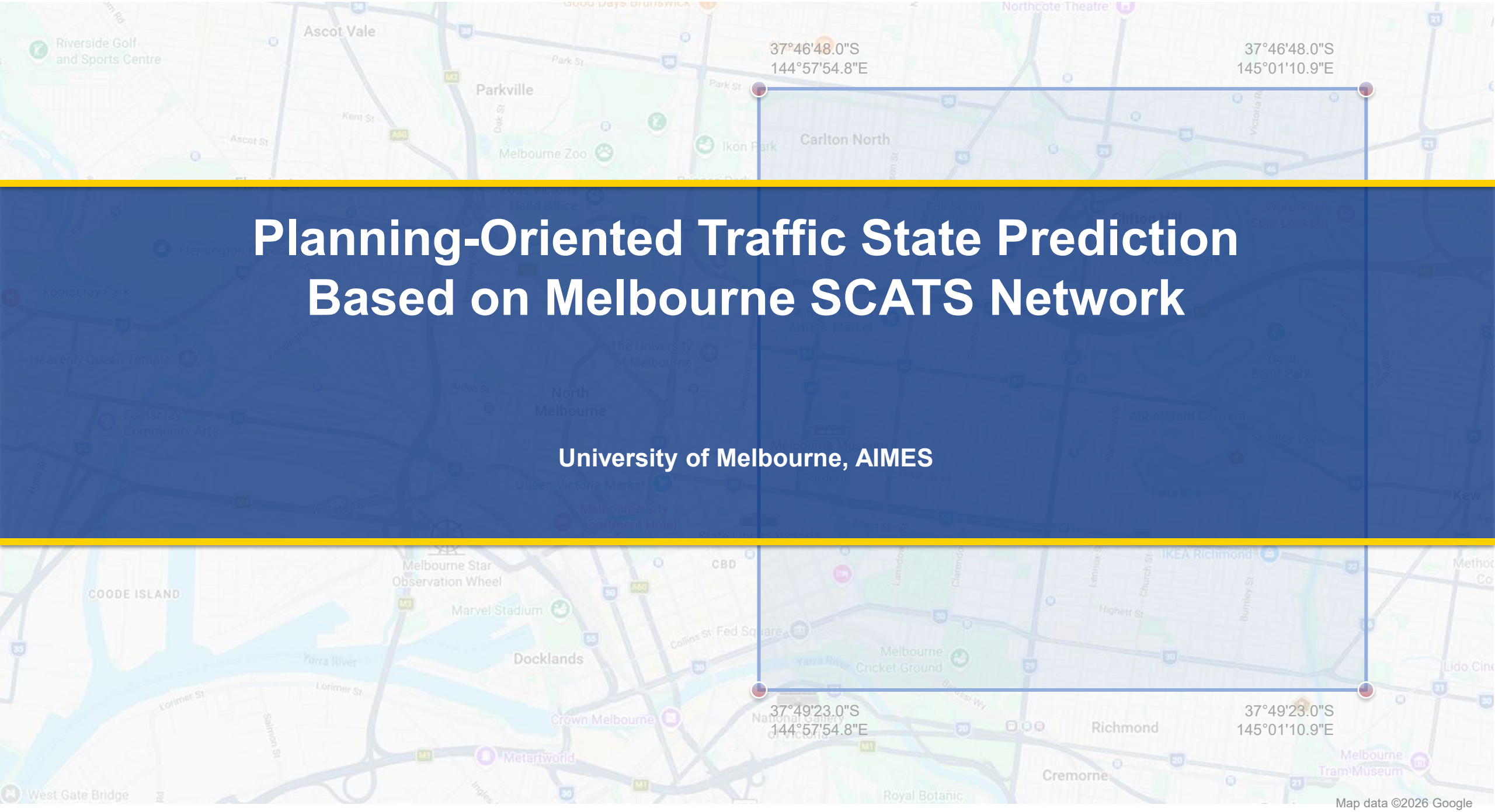


Planning-Oriented Traffic State Prediction Based on Melbourne SCATS Network

University of Melbourne, AIMES

Background and Study Area

Melbourne SCATS signal network — Victoria Open Data Portal

Research objective

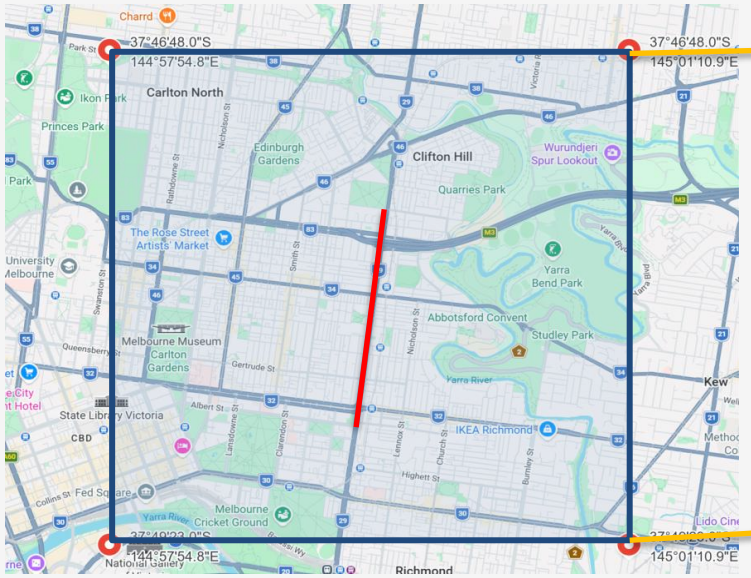
Traffic demand is shaped not only by history, but also by the surrounding built environment visible in satellite imagery.

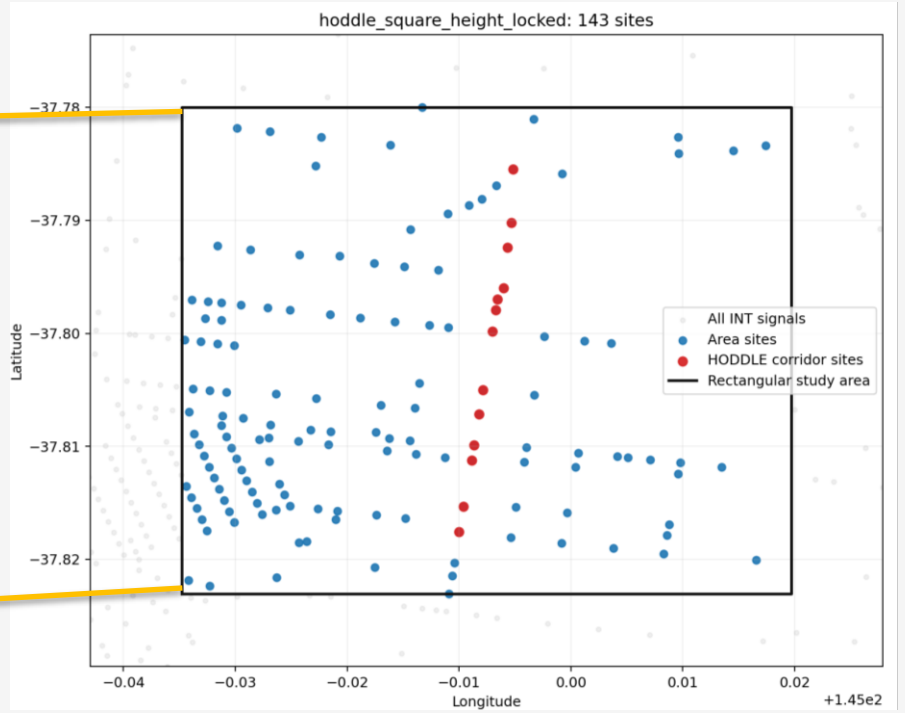
This enables a planning use case: when a site or parcel changes, the updated land-use pattern can be translated into a future traffic-flow estimate.

Current study area

A rectangular study area centered on the Hoddle corridor. (Near Unimelb)
143 SCATS sites; the model uses the 81 with CLUE land-use data.

- Traffic Signal Volume Data — Victorian Open Data Portal
- Victoria Traffic Signals — site coordinates & metadata
- Time range: 2014-2024, 15-min slots (2020 and 2021 left out)
- Satellite imagery: Nearmap, one survey a year





Proposed Multimodal Framework

Route 1 learns visual semantics from satellite images; Route 2 extends with network and traffic topology JSON embeddings.

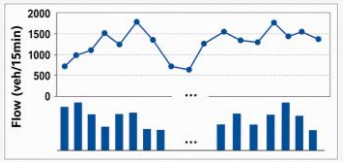
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Inputs

1 Satellite imagery



2 15-min SCATS traffic flow data



- site
- date
- time slot (15-min)
- historical flow

3 Land use information

Site-year alignment / temporal indexing / log-ratio target construction

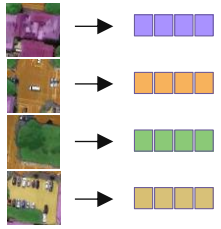
Two-Route Multimodal Modelling

Route 1: Visual Semantic Route

SAM 3.1 semantic region proposal



Per-region DINO embedding



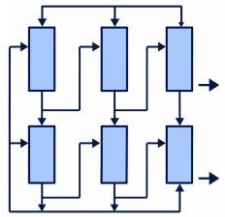
Aggregation by class

1, 2, 3, ⋮, k

Site-year visual feature table

site_id	year	…	f_1	…	f_d
1001	2021	…	…	…	…
1001	2022	…	…	…	…
⋮	⋮	⋱	⋮	⋮	⋮
2050	2024	…	…	…	…

Day-TCN predictor



Target: log ratio to the 2022 historical profile

24-d dynamic temporal context

calendar cyclical · weekend / year · holiday / event · historical base profile

***Route 1** validates and improves the visual frontend;*
***Route 2** adds LLM-derived topology context.*

shared visual frontend

Route 2: LLM-Enhanced Multimodal Route

A Visual embedding — SAM-DINO visual features …

B Network topology JSON — { } JSON

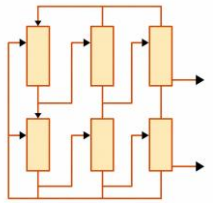
- site_id
- degree
- Neighbors
- edge_weights
- 2-hop summary

LLM embedding: Qwen3-Embedding hidden layer pooling + PCA compression

LLM embedding model → **Net JSON embedding** …

24-d dynamic temporal context

Fusion Day-TCN / multimodal predictor



Outputs

Predicted 15-min traffic flow



Log ratio → volume



Evaluation:
MAE, RMSE, MAPE, GEH, GEH<5%

Route 1 and Route 2

Inputs
Nearmap satellite image
500 m x 500 m per site
CLUE land use
2022 historical profile
Calendar and holidays

Route1
SAM 3: 4 land-cover classes
DINOv3 features, PCA to 16-d
Class aggregation: 135-d
Output: visual feature

Route2
Route1 visual feature
+ network JSON embedding
(Qwen3 embedding, PCA 64)
Output: site-year feature

Predictor
Temporal convolution (TCN)
Imagery, network, land use
24-d calendar & holidays
Ratio to 2022 profile
Output: 96 x 15-min flows

Spring St / Flinders Ln, 2014
CBD: satellite image



Spring St / Flinders Ln, 2014
SAM 3 on the whole 500 m image



Alexandra Pde / Wellington St, 2020
Collingwood: satellite image



Alexandra Pde / Wellington St, 2020
SAM 3 on 150 m tiles



Road Parking / paved Roof Vegetation

Experimental Results

Ablation experiments: test year 2024, 81 intersections, mean of 5 random seeds.

Model	MAE	RMSE	GEH Median	GEH<5 (%)	GEH<10 (%)
Time only	62.22	127.7	4.14	57.2	81.6
Route 1: + imagery	60.55	122.5	3.96	58.8	82.0
Route 2: + imagery + network	**60.14**	**122.3**	**3.93**	**59.1**	**82.3**

Imagery: MAE -1.67 ($p = 0.002$), lower in all 5 seeds.
Road network: MAE -0.41 ($p = 0.38$), not significant.

Land-Use Data: CLUE

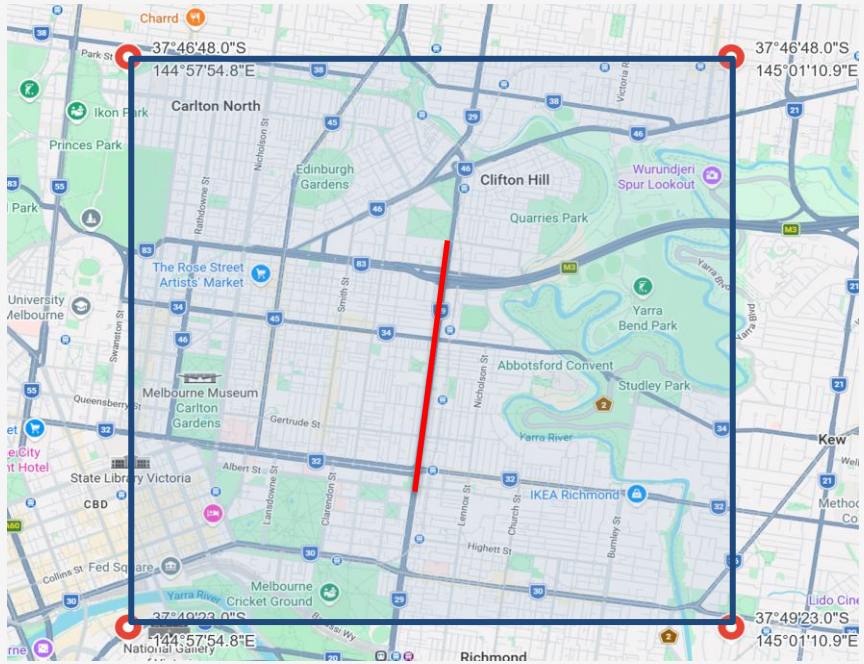
Verify the effectiveness of adding building features. only-time, route1 and route2

Add building feature in traffic prediction

1. Dataset: City of Melbourne: CLUE (Census of Land Use and Employment)
2. Building height data only in City of Melbourne
3. Year coverage: 2002—2024
4. Problem of the satellite image from City of Melbourne: Too much duplicate data, before 2020. (Change another satellite image dateset: Nearmap)

train 2014-2019 + 2022, val 2023, test 2024

Previous work—traffic prediction



2020-2026

2020-2024(5 years, 3 years training)

Land-Use (Building) Features

Add building feature in traffic prediction

43-dim aggregate statistics

Everything inside the site area is summarised into one number per quantity

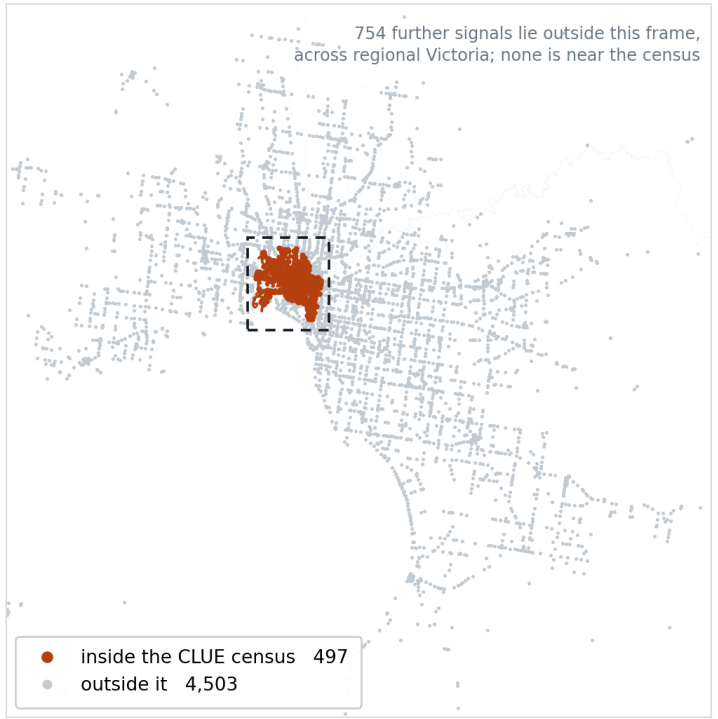
1	Quantity	Buildings
9	Height	floors_mean / max / sum / p90, tall_over_10, tall_over_30, floors_sum_inner, floors_max_inner, floors_sum_weighted
4	Distance	buildings_inner, buildings_weighted, nearest_over_10_m, nearest_over_30_m
26	Land use	share_* × 13, floorshare_* × 13
3	Other	bicycle_spaces, built_last_5y, median_age

Validation in Different Regions

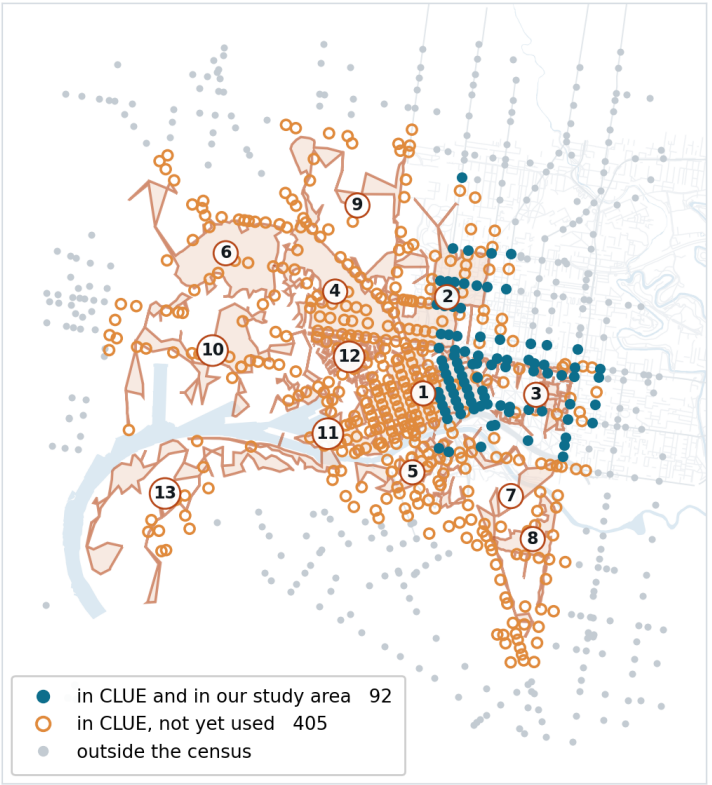
Validate the effectiveness of the model in different regions.

The land-use census reaches one tenth of the signal network

Greater Melbourne — 4,246 of the state's 5,000 SCATS signals

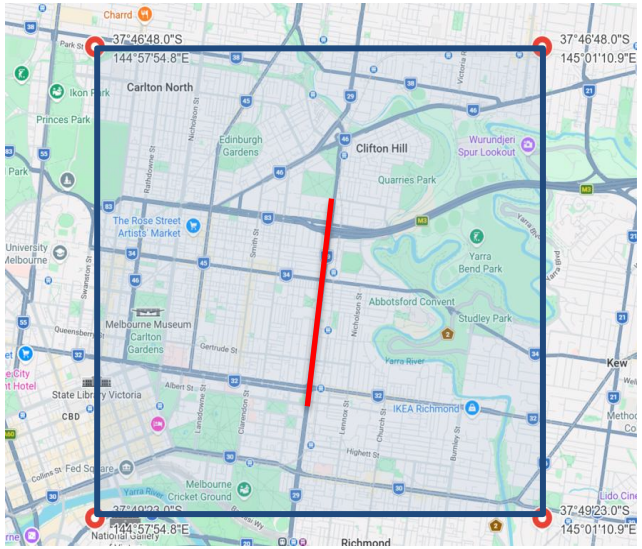


CLUE ∩ SCATS — the 13 small areas and the signals in them



CLUE small area		signals
1	Melbourne (CBD)	116
2	Carlton	60
3	East Melbourne	48
4	North Melbourne	48
5	Southbank	45
6	Kensington	30
7	Melbourne (Remainder)	26
8	South Yarra	26
9	Parkville	24
10	W Melbourne (Industrial)	24
11	Docklands	22
12	W Melbourne (Residential)	16
13	Port Melbourne	12
total		**497**

Previous work



Results: Land Use and a Wider Region

Test year 2024, mean of 5 random seeds (final report, Tables 4.12 and 4.14).

	MAE	GEH_median
Time only	62.22	4.14
Time + land use	61.36	**3.94**
Route 1	60.55	3.96
Route 1 + land use	59.51	**3.88**
Route 1 (371 sites)	59.11	4.13
Route 1 + land use (371 sites)	59.25	**4.11**
Route 2	60.14	3.93
Route 2 + land use	59.18	**3.84**
Route 2 (371 sites)	59.24	4.13
Route 2 + land use (371 sites)	58.76	**4.11**

Land-Use Sensitivity Analysis

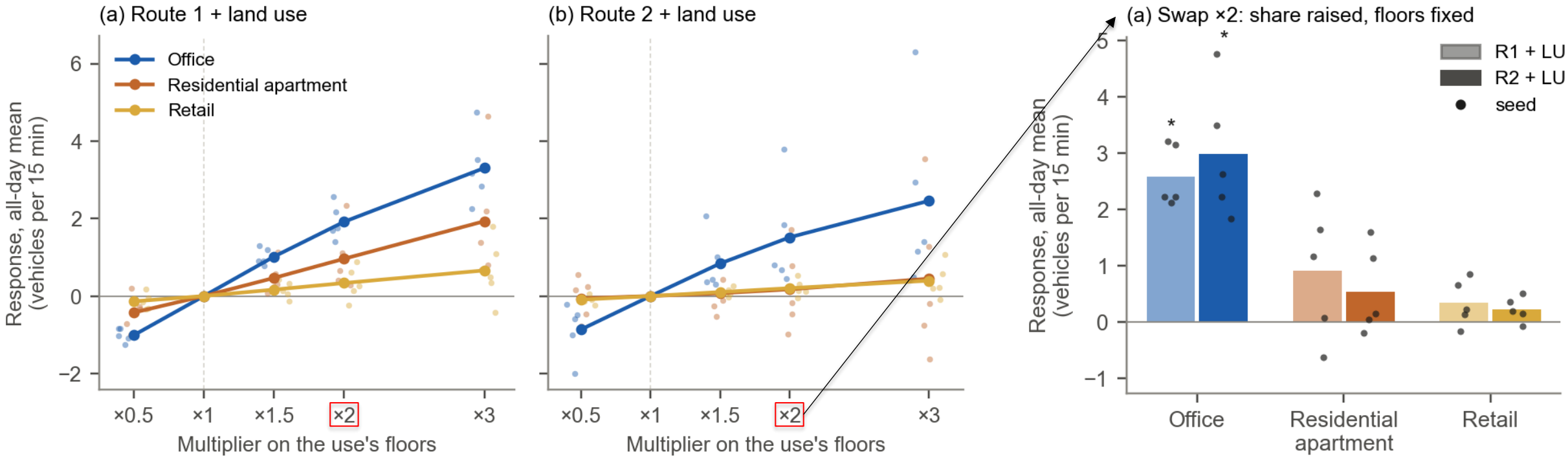
Sensitivity analysis of building attribute features.

New building

For example: office space from 10->20
Then total space from 50->60

Change building attribute from other 12 types buildings

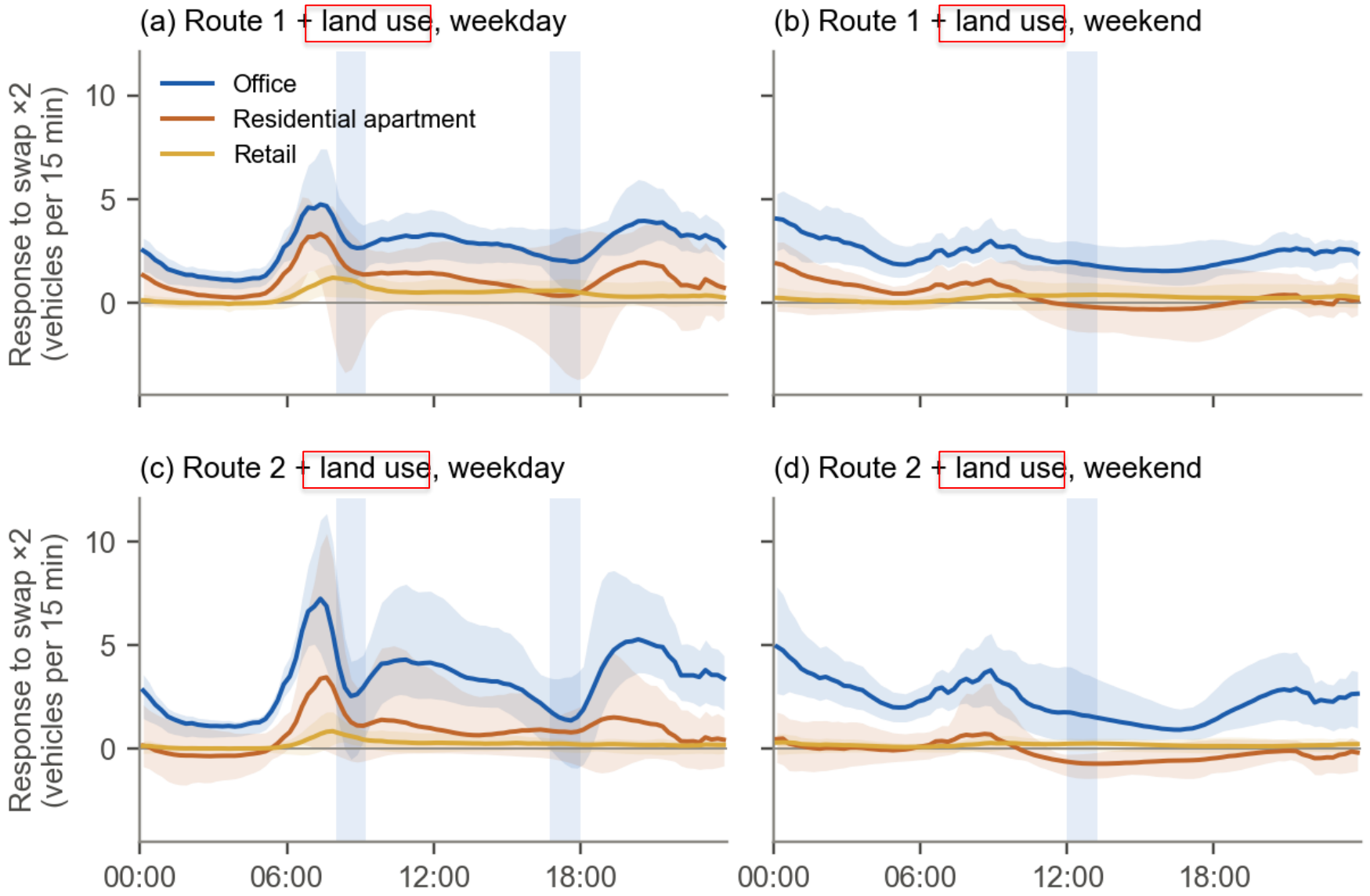
For example: office space from 10->20
Then total space remains 50



Land-Use Sensitivity: Time of Day

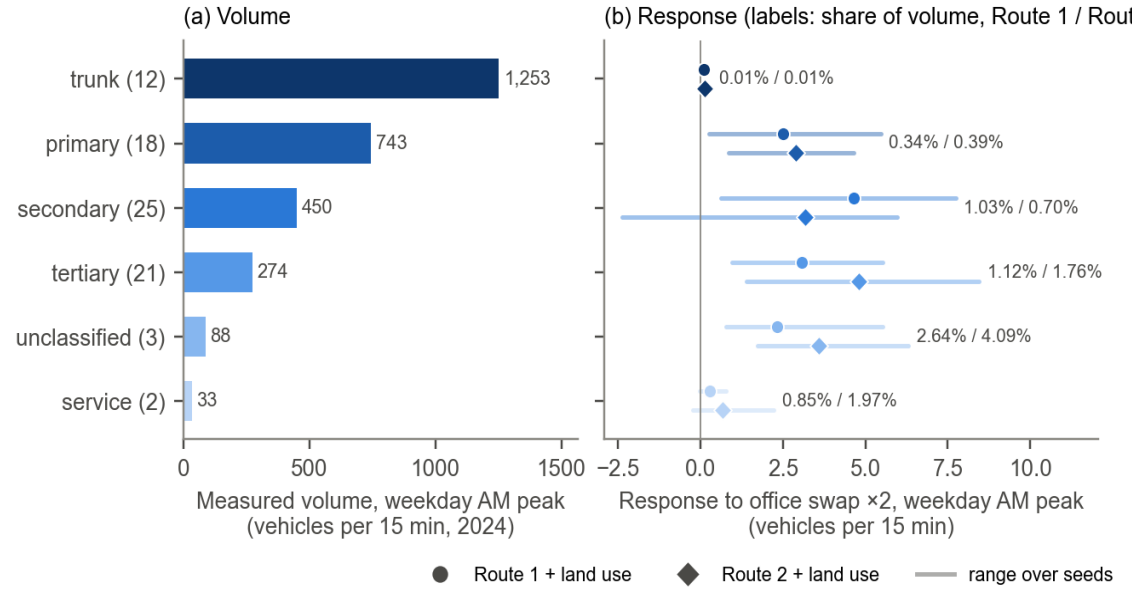
Sensitivity analysis of building attribute features.

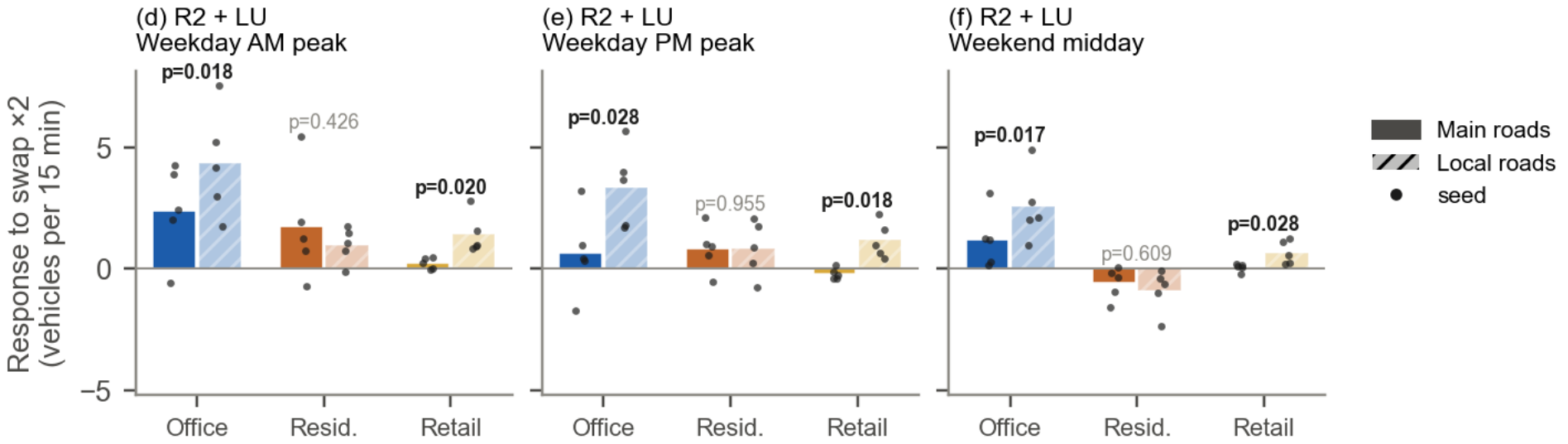
Swap
Floorspace * 2



Land-Use Sensitivity: Road Class

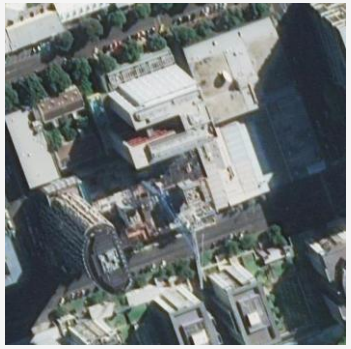
Sensitivity analysis of “all roads” and “only main roads”

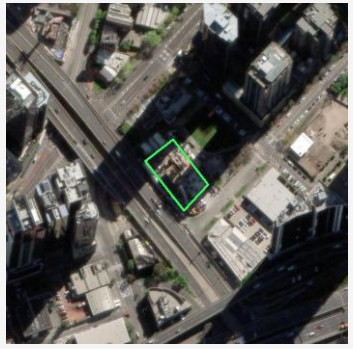
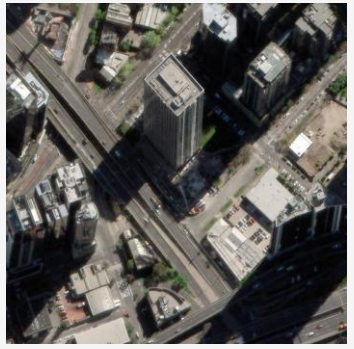




Development Scenario Imagery: 3D Model + LLM

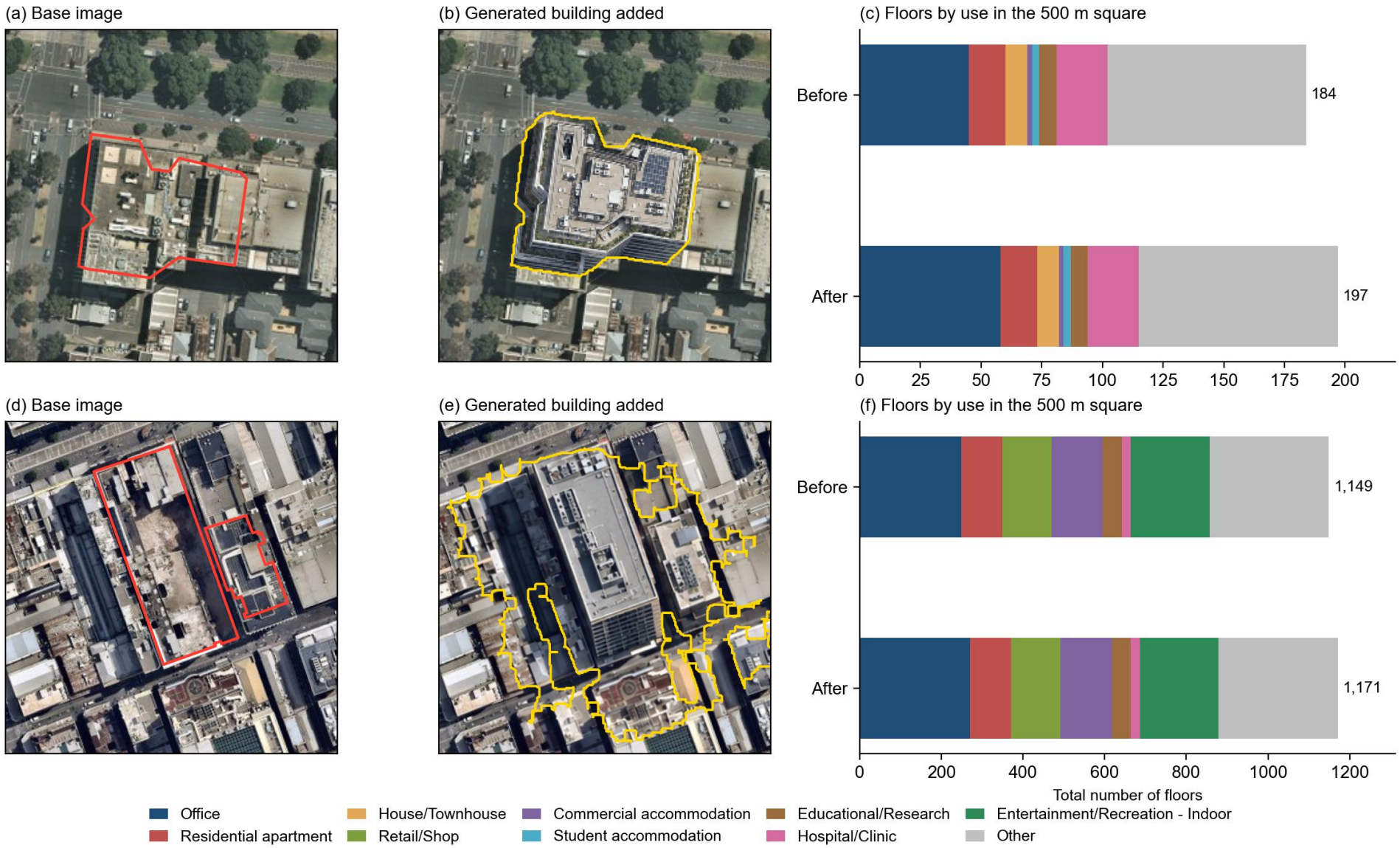
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Before	Generate	Real
		
		
		

Before	Generate	Real
		
		
		

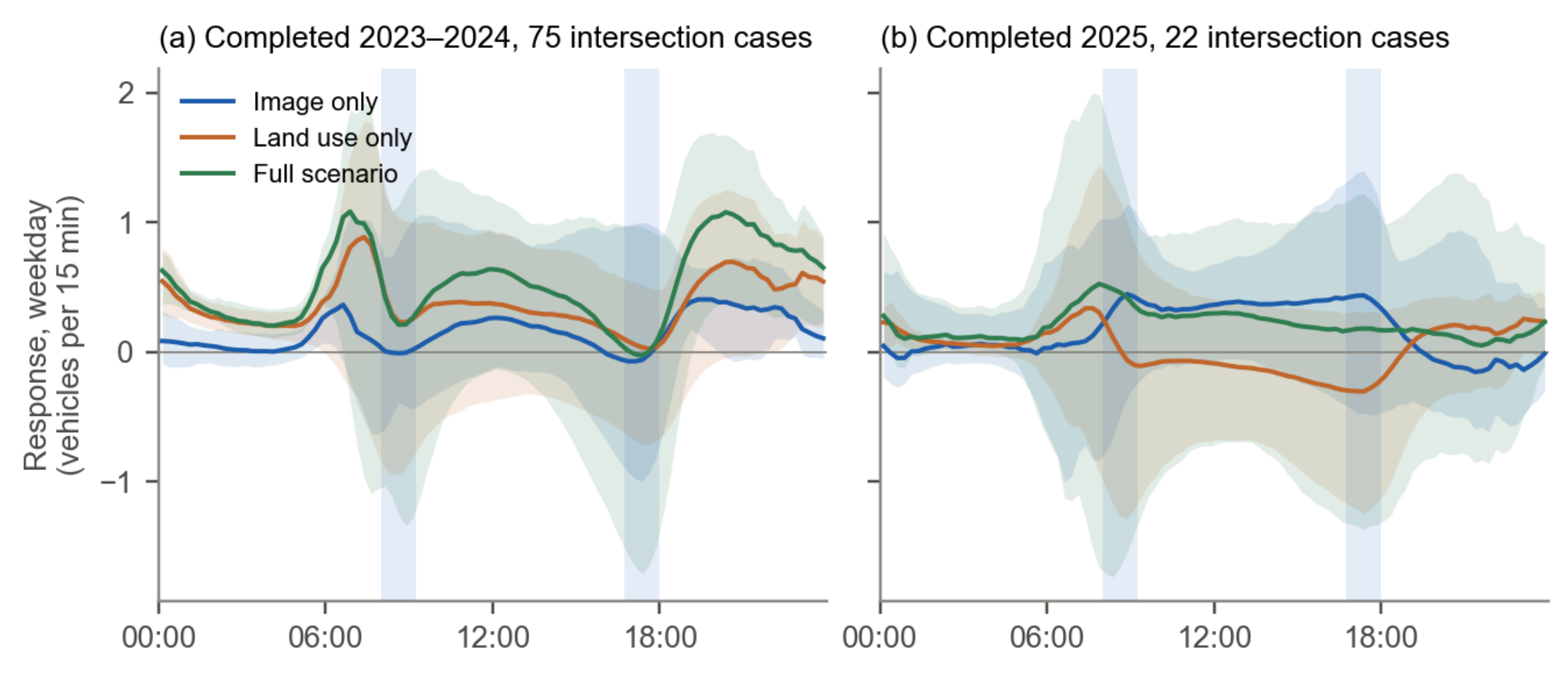
Generated Imagery: Test Cases

Real developments completed in 2023-2025, generated on imagery from before construction.



Generated Imagery in Traffic Prediction

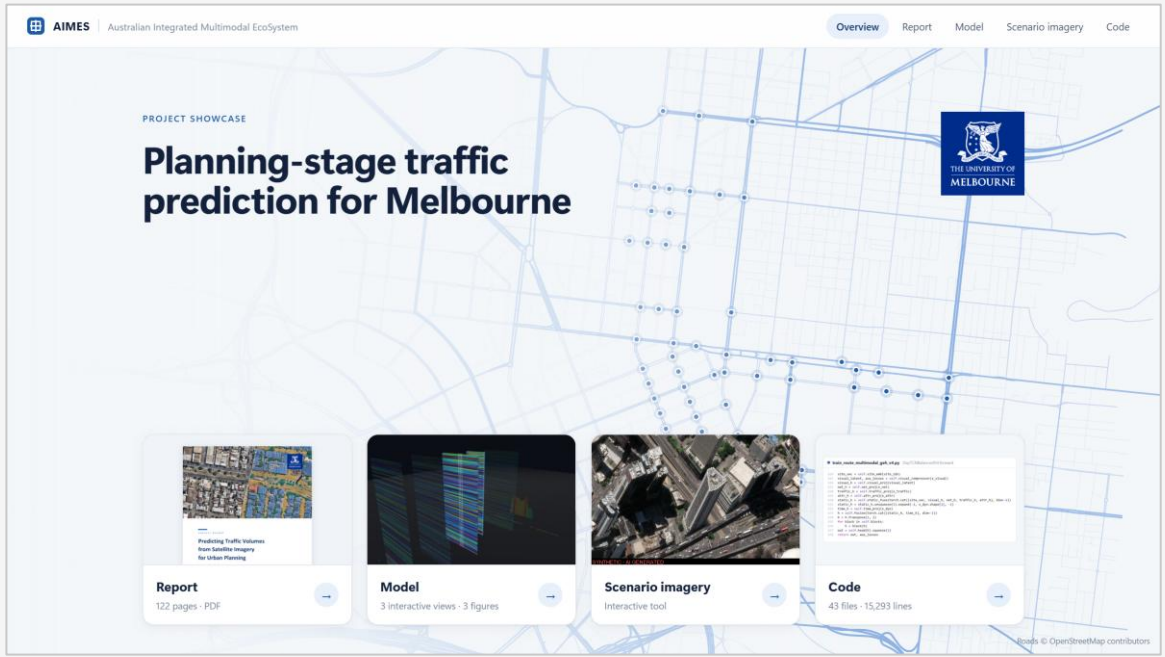
Response of Route 2 + land use to each case against no development (weekdays, 5 seeds).

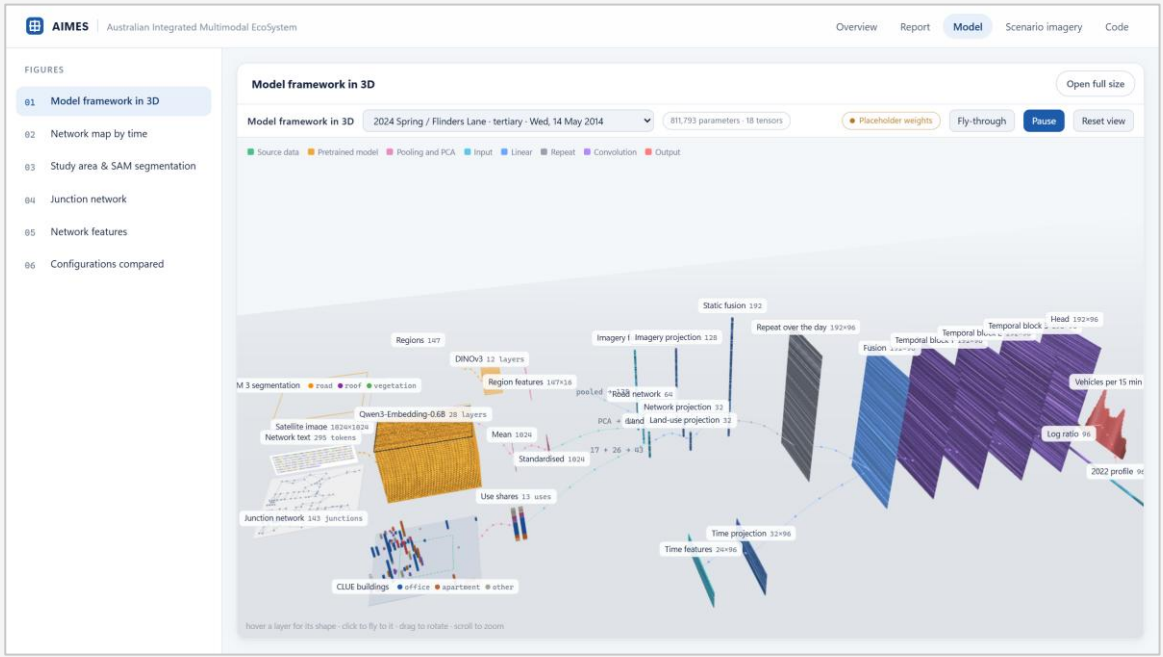


Accuracy with generated imagery: median GEH 3.50-4.06, MAE changes by at most 0.15 veh/15 min.

Full scenario (generated building and updated land use): +0.49 and +0.19 veh/15 min, not statistically significant.

Project Website





Project website: cjsrc.github.io/Melbourne_traffic_showcase

Scenario imagery tool: cjsrc.github.io/Melbourne_traffic_showcase/tool.html

Report (122-page PDF), interactive model views, scenario imagery examples and the project code.